

Raju

System Test Engineer

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Automotive **System Test Engineer** with over **4** years of experience in the automotive industry. Expertise in developing and executing test plans for advanced automotive systems, including **Infotainment, ADAS & Telematics** ECUs. Skilled in system integration, test automation, and troubleshooting of electronic control units (ECUs) for Vehicles. Proven track record of ensuring system performance, meeting industry standards and delivering effective test solutions in fast - paced environments.

Experience

- & Hands-on experience with different **ECUs** such as **Infotainment, Telematics** and **ADAS** systems, Performing both system-level and bench level testing to ensure optimal performance.
- & Proficient in utilizing **CANOE** tool features, including **Trace window, Graphical window, Panel window** and **IG block**, and also having experience in Canalyzer.
- & Experienced in testing and validating automotive communication protocols such as **CAN, CANFD, CANTP, UDS** and Automotive Ethernet.
- & Proficient in Verification and Validation (**V&V**) processes throughout the Software Development Life Cycle (**SDLC**), ensuring robust quality assurance and adherence to industry standards.
- & Demonstrated expertise in test execution, reporting, and defect tracking, with thorough knowledge of the testing life cycle and hands-on experience using **JIRA**.
- & Experienced in designing test cases to meet business and system requirements, ensuring full traceability via **IBM Rational DOORS**.
- & Experienced in Requirements analysis and test case preparation, utilizing testing techniques such as **Boundary Value Analysis, Equivalence partitioning, Use Case testing**, and **Error Guessing** to ensure comprehensive test coverage and optimal software quality.
- & Proficient in reading Diagnostic Trouble Codes (**DTCs**) and understanding Negative Response Codes (**NRCs**)
- & Strong expertise in **AUTOSAR, HIL** test environments, and adherence to **ASPICE** and **ISO 26262** standards for functional safety.
- & Developed and executed complex test cases using **CAPL** scripting in **Canoe** to automate test scenarios for Infotainment, ADAS and cluster systems, significantly reducing manual testing time and improving efficiency.
- & Proficient in utilizing industry-leading testing tools and methodologies, including **Smoke, Functional, Regression, Sanity** and Exploratory testing, to ensure product quality and safety compliance.
- & In-depth knowledge of **UDS** protocol and working with service IDs (e.g., **10, 11, 22, 2E, 3E, 27, 31, 19, 28, 85, 14**) for effective communication and diagnostics with ECUs.
- & Hands-on experience with **Dyna4** for automating tests and analyzing data, leading to more efficient test processes and accurate performance assessments.

EMPLOYMENT HISTORY

System Test Engineer with **4** years of experience at **Jasmin Infotech**, Chennai - (600100)

From **March - 2022** to present.

KEY SKILLS

- & **Communication Protocols** : CAN,CAN-TP, UDS, CANFD, ETHERNET.
- & **Testing Tools** : Canoe, Canalyzer, Dyna4 Diag tool.
- & **Bug Tracking Tool** : JIRA, ALM
- & **Methodology**: V&V, ASPICE.
- & **ISO Protocols**: ISO 26262.
- & **Other Skills** : Requirement Analysis, Log Analysis Test Report Preparation, Software Model Verification & Validation
- **Flashing Tools**: USB Flashing, UDS Flashing, ADB Flashing, Renesas Debugger, Fast boot flashing.
- **Hardware Tools**: GPS Simulator, Frequency generator Vector Box-VN5610A.
- **HIL Bench**: VT STUDIO
- **Software Tool**: TeraTerm, PUTTY
- **Management Tool**: IBM Rational Doors
- **Programming Language**: CAPL Scripting and exposure in Python.

PROJECTS

Project 1: GM's Vision X **Client:** General Motors
February - 2024 to Till Date

Environment: INFOTAINMENT & Telematics

Responsibilities:

- & Extensive experience in testing, validating and ensuring functionality of automotive infotainment systems, including **Navigation, Media connectivity, Voice Recognition, Android Auto & Car Play**.
- & Expertise in flashing software in the infotainment system using **USB, Fast Boot Flashing** and **Renesas Debugger**.
- & Experience in validating Bluetooth profiles including **GAP, MAP, SAP, HFP, AVRCP, ADP**, and understanding Bluetooth protocol stack.
- & Strong Knowledge of **Test Bench setup** and **Test harness** experience to simulate real-world conditions for precise testing.
- & Extracted and analyzed **serial logs** using **TeraTerm Tool, CAN logs**, and **USB logs** for Debugging and resolution.
- & Tested Human-Machine Interface (**HMI**) features, such as **AC controls, Blower speed**, and **Seat functions**, using **Canoe** tool for signal simulation.
- & Participated in testing Navigation features including **Point of Interest, History, Paved Roads**, and **Unpaved Roads**, ensuring precise and reliable capabilities.
- & Executed In-depth **Radio functionality testing**, Verifying **AM/FM** reception, **Tuning**, and **Display functionality**.
- & Strong understanding of the **telematics** architecture, including multiple OSI layers relevant to vehicle connectivity.
- & **Performed diagnostic validation** of **Infotainment systems**, ensuring proper functionality across features like **Bluetooth, Navigation, Audio**, and **Display**.
- & **Identified** and **analyzed Diagnostic Trouble Codes (DTCs)** associated with **Infotainment system faults**, such as **communication issues** and **hardware failures**.
- & **Hands-on expertise** in features like **eCall, bCall, door open alerts, vehicle charging history monitoring, remote analog and digital logging, remote lighting control, remote climate control**, and **seat belt status monitoring**.
- & Skilled in Flashing Telematics ECUs using **self Fota** and **Remote-Fota** Methods.
- & **Monitored** and **validated the diagnostic functionality** of features such as **Android Auto, Apple Car Play, Bluetooth**, and **Navigation**.
- & Knowledgeable in **OTA (Over-the-Air) software update procedures for Telematics ECUs**.
- & Experienced in conducting diagnostic performance testing and DTC (Diagnostic Trouble Code) validations in **Telematics ECUs**.
- & Verified processes related to **software download, installation, and validation** within telematics systems.

Client: Skoda

Environment: ADAS

- & Testing **Forward Collision warning** and **Parking assistance system** has been a part of my experience.
- & Analyzed system requirements for **Skoda's Forward Collision Avoidance System**, authored comprehensive test cases, and validated system functionality to meet specified performance standards.
- & Managed and tested **Ultrasonic sensor** systems for **Forward** and **Backward Collision Warning**, ensuring accurate detection, response times, and system reliability in real - world driving scenarios.
- & Proficient in how Ultrasonic sensors communicate with **ADAS** ECUs, including signal sending and data integration from various ranges (**5m, 3m, 1m**).
- & Extensive experience testing **ADAS** features such as **MVC, RVC, ACC**, and **Ultrasonic parking sensors**, Ensuring quality performance and functionality.
- & Experienced in using **AVB panels, DYNA 4**, and **Canoe** tools for signal transmission and test execution in **Infotainment** and **ADAS** systems.
- & Performed **HIL** bench simulation with **DYNA 4** tool to validate **ADAS** ECU functionality, providing real- time feedback and **3D visualization** for comprehensive testing.
- & Conducted closed-loop testing in **VT-Studio HIL setup**, ensuring real -time feedback and system validation under varied driving conditions.
- & In-depth knowledge in validating **Anti-lock Braking System (ABS)** functionalities via **HIL bench testing**, ensuring reliable performance under various conditions.
- & Hands-on experience modifying the **Gateway Module's** communications format from **CAN** to **Ethernet**, facilitating seamless data transmission to the **ADAS ECU**.
- & Extensive knowledge and practical application of industry tools such as **Canoe, DYNA 4, JIRA, IBM Rational Doors**, for test case execution, defect management, and requirement analysis.
- & Led defect tracing efforts, identifying root causes for system malfunctions, and implemented corrective actions, improving system stability and overall **ADAS** performance.
- & **Developed** and **implemented fault creation procedures**, including **missing messages, invalid signals, over-temperatures**, and **over-voltages** for **ADAS** systems.
- & **Prepared** and **executed diagnostic test cases** for **function-level** based on system specifications and diagnostic requirements.

EDUCATION

- Completed Bachelor of technology in Computer Science Engineering

Declaration:

- & I hereby declared that all the above mentioned information is true to the best of my knowledge and belief

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